CrossPiezo: Elite-Tail Instability in Cross-Database Piezoelectric Screening

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Abstract

High-throughput materials screening assumes that top-ranked candidates are reproducible across databases, yet agreement is usually reported as a single global correlation. We introduce CrossPiezo, a structure-matched benchmark of 573 materials shared by the Materials Project and JARVIS, and quantify screening resolution with three coordinate-invariant piezoelectric response scalars. Cross-database concordance is nearly absent in the elite 1–10% candidate range and increases only after substantially broadening the screened pool; the tighter 207-pair panel shows the same pattern. In contrast, unit-cell volume and band gap are markededly more reproducible across the same sources, indicating that response properties have substantially higher workflow sensitivity. These results show that aggregate correlation can overstate the reproducible resolution of high-throughput piezoelectric screening.

1 Introduction

Piezoelectric response tensors are central to electromechanical materials design, yet costly to compute with density-functional perturbation theory (DFPT) [1, 2]. The Materials Project (MP) and JARVIS projects have released thousands of computed tensors [3, 4], and recent equivariant neural networks report strong in-source predictive agreement [5, 6, 7, 8, 9]. Improved agreement with a held-out subset of one database does not guarantee that a candidate is robustly ranked across independently generated high-throughput datasets [10, 11].

We introduce *screening resolution* as a diagnostic lens. Rather than asking "what is the average rank correlation between two sources?" we ask "at what elite fraction q does the overlap between source-specific top-q sets become statistically distinguishable from chance?" This distinction matters for materials discovery: a model validated only on aggregate correlation may still disagree with an independent source on the very candidates an experimenter would pursue [12, 13, 9].

This paper makes three contributions. First, we release a frozen, structure-matched benchmark of 573 MP–JARVIS pairs and three coordinate-invariant response scalars. Second, we show that cross-database concordance is resolution-dependent: near-chance in the elite tail, weak in the intermediate band, and only modest in the broad band. Third, we demonstrate that scalar controls are far more consistent than the piezoelectric response, and that a simple source-robust portfolio raises worst-source recall by +0.386 (95% CI [0.214, 0.500]) compared with the better single source, providing a practical risk-management tool while remaining no substitute for independent validation.

2 CrossPiezo benchmark

The CrossPiezo-Invariant-v1 panel contains P0 ($n = 573$) and P2 ($n = 207$) structure-matched pairs with identical or near-identical relaxed crystal structures across MP and JARVIS [14, 15]. Figure 1a summarises the matching pipeline: 8,316 processed records were reduced to 1,266 structure-level overlaps, then to the frozen P0 and P2 panels. Panel membership was fixed after a response-blind manual audit of the top-matched candidates using a frozen rule set; all audit decisions are recorded in the repository.

For each source we compute three coordinate-invariant scalars from the piezoelectric stress tensor $\mathbf{e}$ (Figure 1b): F1 = Cartesian Frobenius norm $\|\mathbf{e}\|_F$; F3 = true collinear longitudinal maximum $\max_{\|\boldsymbol{n}\|=1} |n_i e_{ijk} n_j n_k|$; F4 = Kelvin/Mandel operator norm on symmetric strain. F3 isolates the longitudinal single-crystal response; F1 and F4 are more relevant for polycrystalline or norm-based screening.

Screening resolution is defined by sweeping the screened quantile q from 1% to 50% and comparing the source-specific top-q sets (Figure 1c). At each q we report the chance-adjusted Jaccard overlap $\widetilde{J}_q$ and a simultaneous 95% confidence band; the normalised area under this curve is denoted nAUCC (see Methods). The key diagnostic is whether concordance is reliably sustained as the candidate pool narrows from broad (20–50%) to elite (1–10%).

3 Elite-tail screening resolution

Figure 2 shows the flagship screening-resolution results. On P0 the three functionals behave similarly: full-range nAUCC is 0.105 for F1, 0.110 for F3 and 0.106 for F4 (Table 1). The simultaneous lower confidence bound first exceeds 0 at $q = 33\%$ for F1, but a five-quantile persistent onset at $\delta = 0.05$ is not observed. P2 is even weaker: F1 nAUCC = 0.060, F3 nAUCC = 0.043, F4 nAUCC = 0.046. Thus, even when half of the candidate pool is screened, the two sources barely agree more than would be expected by chance.

The banded partial nAUCC summary (Figure 2, bottom row and Table 1) makes the resolution-dependence explicit. On P0 F1, concordance is essentially absent in the elite band (partial nAUCC = 0.003), modest in the intermediate band (0.076), and only clearly positive in the broad band (0.145). The same graded pattern holds for F3 and F4 and is preserved, though attenuated, on P2.

Corrected high-response sensitivity analyses avoid the pooled-selection collider bias (Supplementary Table S1). The dual-high subset ($f = 0.10$, $N = 58$) has nAUCC = -0.046, and the JARVIS- and MP-anchor subsets give consistently weak or negative elite-tail concordance. This confirms that the elite-tail resolution gap is not an artifact of the high-response definition.

Table 1: Screening-resolution summary. Full-range nAUCC and partial nAUCC by band for P0 ($n = 573$) and P2 ($n = 207$). $q_{0.05}^{\text{persist}}$ requires five consecutive lower-confidence bounds above 0.05.

Panel	Metric	nAUCC	Elite	Intermediate	Broad	$q_{0.05}^{\text{persist}}$
P0	Frobenius norm	0.105	0.003	0.076	0.145	–
P0	Longitudinal	0.110	-0.000	0.061	0.160	38%
P0	Kelvin	0.106	-0.004	0.069	0.151	42%
P2	Frobenius norm	0.060	-0.023	0.030	0.095	–
P2	Longitudinal	0.043	-0.021	-0.037	0.089	–
P2	Kelvin	0.046	-0.021	-0.024	0.089	–

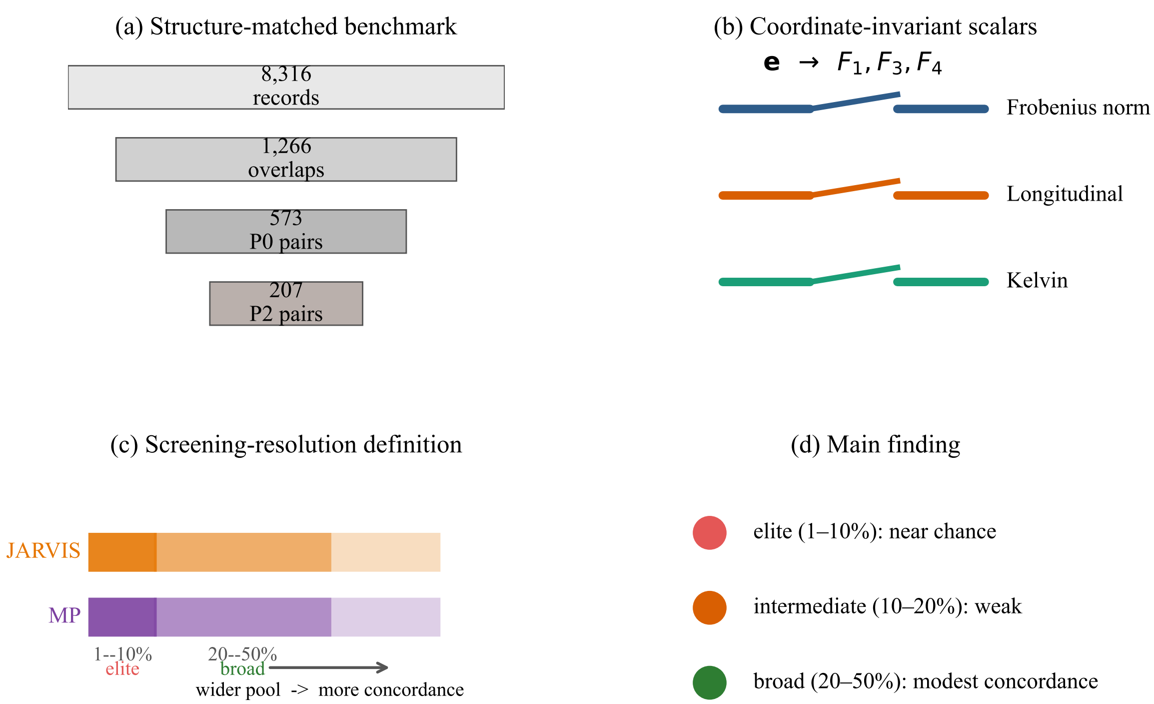

Figure 1: CrossPiezo benchmark concept. (a) Structure-matched data pipeline. (b) The three coordinate-invariant piezoelectric response scalars computed from the stress tensor $\mathbf{e}$. (c) Screening-resolution compares source-specific top-q sets; concordance generally grows as the screened pool broadens. (d) Main finding: cross-database concordance is near-chance in the elite tail and only becomes modest in the broad band.

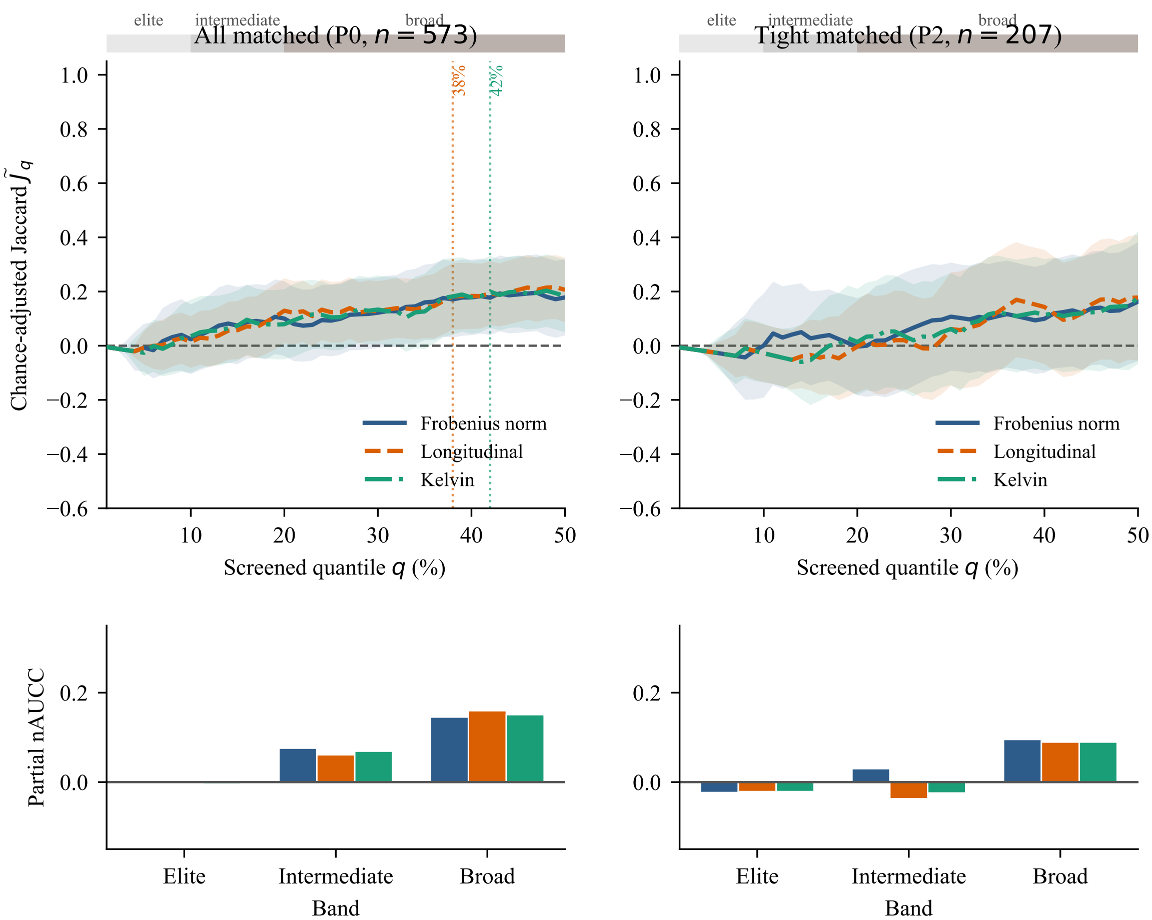

Figure 2: Flagship screening-resolution results. Top row: chance-adjusted Jaccard overlap $\widetilde{J}_q$ for P0 (left) and P2 (right), with simultaneous 95% confidence bands. Line styles distinguish the three invariants: Frobenius norm (solid), Longitudinal (dashed), Kelvin (dash-dot). Top colour bars mark the elite (1–10%), intermediate (10–20%) and broad (20–50%) bands. Bottom row: partial nAUCC by band, showing that concordance is weakest in the elite band and grows only after broadening the candidate pool.

4 Response properties are unusually workflow-sensitive

If the elite-tail gap were a generic failure of structure matching, all properties should show similarly weak cross-source concordance. They do not. Figure 3 compares piezoelectric F1 against unit-cell volume, band gap, energy above hull and dielectric trace. Volume is the strongest control (P0 Kendall $\tau = 0.967$, nAUCC = 0.918), followed by band gap ($\tau = 0.908$, nAUCC = 0.897). All controls have positive $\Delta\tau$ and higher nAUCC than F1; the energy-above-hull control is flagged as a same-field copy because many stable materials share a zero hull value in both sources (open circle in Figure 3).

This contrast indicates that piezoelectric response has higher workflow sensitivity in these two sources. It does not prove that response disagreement is mainly conventional; an irreducible physical component cannot be ruled out without an independent third protocol or experiment.

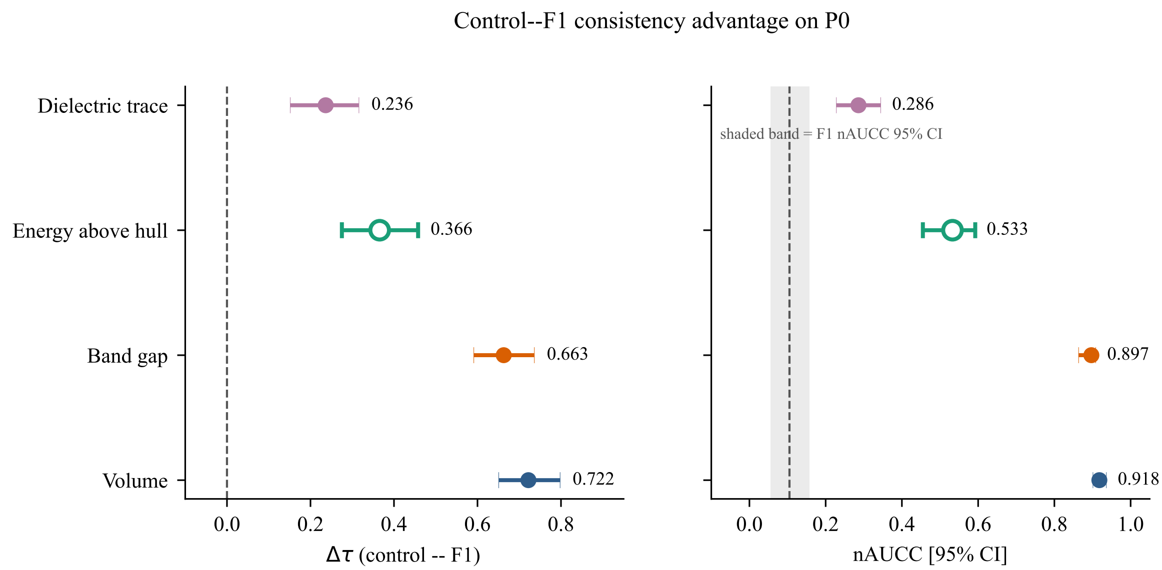

Figure 3: Property-control comparison on P0. Left: advantage of each control over piezoelectric F1 in Kendall τ, with 95% confidence intervals. Right: absolute nAUCC for each control, with the dashed line and shaded band showing the F1 nAUCC point estimate and 95% CI. All controls are substantially more cross-source consistent than the piezoelectric response. The energy-above-hull control is marked with an open circle because many stable materials share a zero hull value in both sources.

5 Discussion

The practical consequence of the resolution gap is that single-source elite lists are not automatically reproducible. Supplementary Figures S2–S3 and Tables S3–S4 illustrate the resulting candidate-selection ambiguity and report a source-robust portfolio analysis. On P0 F1 at $q^* = 10\%$, the balanced-union strategy achieves a worst-source recall of 0.526 and improves over the better single source by +0.386 (95% CI [0.214, 0.500]) (material-level paired difference with a group-bootstrap 95% CI). This shows that aggregation can materially improve worst-case coverage, but it does not validate physical truth.

5.1 Screening resolution versus global rank correlation

The weak elite-tail resolution is a statement about the reproducible resolution of these two databases, not about the physical correctness of either source. A model can achieve strong aggregate agreement on a held-out subset of one database while still disagreeing with an independent source on the specific candidates a human experimenter would pursue. We recommend that future high-throughput screening reports screening-resolution curves, chance-adjusted overlaps, banded partial nAUCC and source-robust portfolio diagnostics alongside aggregate correlations and global rank correlations.

5.2 Why response properties are more workflow-sensitive

Simpler scalar observables such as unit-cell volume and band gap are reproduced far more reliably across MP and JARVIS than the piezoelectric tensor. This pattern is consistent with higher workflow sensitivity for response tensors: piezoelectric constants depend on density-functional settings, strain conventions, finite-difference or DFPT details, and post-processing choices that are not fully aligned between the two databases. The control contrast does not establish that conventions dominate; an irreducible physical disagreement cannot be ruled out without a third independent protocol or experiment.

5.3 Consequences for AI-guided materials selection

For AI-guided discovery, our results imply that a model trained and validated on a single high-throughput source may inherit that source's idiosyncrasies. Elite candidates selected from a single source should not automatically be treated as robustly high-performing. Cross-source rank aggregation can improve worst-case coverage: on P0 F1 the balanced-union strategy raises worst-source recall from 0.14 for either single source to 0.526, a material-level paired gain of +0.386 (95% CI [0.214, 0.500]). This is a risk-management strategy, not a substitute for independent validation.

5.4 Limits of two-source evidence

Our conclusions are source-pair specific. Only MP and JARVIS are compared, and full tensor convention alignment was not performed. Elite-tail samples are small ($\sim$6 materials at 1% in P0, $\sim$2 in P2), so elite-tail statements are lower-bound diagnostics rather than precise population estimates. A 48-material third-protocol adjudication plan has been pre-registered to provide independent computational validation; experimental adjudication would be required to settle source correctness on specific candidates [16, 11].

6 Methods

Paired universe. P0 and P2 were built by structure matching of relaxed CIFs followed by a response-blind manual audit of the top candidates using a frozen rule set; full details and the frozen membership file are released in the repository. At the 1% elite quantile the effective sample is approximately 6 materials in P0 and 2 materials in P2.

Response functionals. For each source we compute three coordinate-invariant scalars from the piezoelectric stress tensor $\mathbf{e}$ [17]: F1 = $\|\mathbf{e}\|_F$, F3 = $\max_{\|\boldsymbol{n}\|=1} |n_i e_{ijk} n_j n_k|$, and F4 = Kelvin/Mandel operator norm on symmetric strain.

Chance-adjusted concordance curve. For each panel, metric and quantile $q = 1\%, \ldots, 50\%$, we report the observed top-q Jaccard overlap J_q, the exact hypergeometric null expectation $\mathbb{E}[J_q]$ under independent rankings, and the chance-adjusted overlap

$$\widetilde{J}_q = \frac{J_q - \mathbb{E}[J_q]}{1 - \mathbb{E}[J_q]}. \tag{1}$$

We construct a simultaneous 95% confidence band around $\widetilde{J}_q$ using a paired bootstrap with a studentized sup-norm critical value [18, 19]. The normalised area under the concordance curve is

$$\mathrm{nAUCC} = \frac{1}{q_{\max} - q_{\min}} \int_{q_{\min}}^{q_{\max}} \widetilde{J}_q \, dq, \tag{2}$$

and the persistent consensus onset is the smallest q for which the lower band exceeds a threshold δ for at least five consecutive quantiles. nAUCC is a descriptive summary; uncertainty is conveyed by the simultaneous bands on the underlying curve.

Banded reporting. The 1–50% curve is split into elite (1–10%), intermediate (10–20%) and broad (20–50%) bands; partial nAUCC and simultaneous bands are reported for each.

Corrected high-response sensitivity. Phase 7C uses three corrected definitions: (i) `dual_high`: both source values exceed the $1 - f$ quantile of $\min(F_J, F_M)$; (ii) `jarvis_anchor`: top f by JARVIS, evaluated on MP; (iii) `mp_anchor`: top f by MP, evaluated on JARVIS. The legacy pooled selection is retained only as an exploratory check with a conditional permutation.

Controls and provenance. We compare piezoelectric response against volume, band gap, energy above hull and dielectric trace. For each attribute we record source field, source database, unit, calculation type/version, missingness and a flag for possible same-field copying (more than 5% identical JARVIS/MP values) [16]. We report $\Delta\tau = \tau_{\text{control}} - \tau_{\text{piezo}}$ and ΔnAUCC with grouped paired-bootstrap confidence intervals, permutation p-values and Benjamini–Hochberg FDR [20].

Robust portfolio benchmark. We fix an elite fraction $q^* = 10\%$ and evaluate strategies on the full panel at budget factors $b = 1.0$, 1.5 and 2.0. Strategies include single-source, average percentile, Borda count, maximin percentile, intersection-first, balanced union, disagreement abstention and a greedy minimax oracle [21, 22]. The primary summary metric is worst-source recall (material-level). Paired differences versus the better single-source baseline are computed as material-level worst-source recall differences using a grouped bootstrap that resamples reduced-formula groups with replacement and projects the global selection onto each bootstrap sample (see Supplementary Note 1). 5-fold grouped cross-evaluation is also performed; CV estimates should be interpreted as an optimistic upper bound because different structural entries of the same reduced formula may leak across folds.

7 Data availability

Hash-bound result tables, panel membership, the Phase 7C manifest, the Phase 8A audit reports, the manuscript-numbers JSON and reproducibility instructions are released in the CrossPiezo repository. A permanent archive containing the frozen manifest, all CSV artifacts, figures and the figure-generation script has been prepared for Zenodo/Figshare deposit; a private reviewer link will be provided before submission.

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